

Finding Zeros of Quadratic Functions Graphically

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A **zero** (or **x -intercept**) of a quadratic function is the value of x where the graph crosses or touches the x -axis.

Zeros of a Quadratic Function

The **zeros** are the solutions of the equation obtained by setting $y = 0$.

$$y = 0$$

The zeros are the x -values where the graph intersects the x -axis.

Factored Form

A quadratic function in **factored form**

$$y = a(x - r_1)(x - r_2)$$

has zeros at $x = r_1$ and $x = r_2$.

Standard Form

A quadratic function in **standard form**

$$y = ax^2 + bx + c$$

may require graphing first before identifying its zeros.

Two Real Zeros

If the graph crosses the x -axis at two points,
the function has **two real zeros**.

Two intersections

$\Rightarrow$ two real zeros

One Repeated Real Zero

If the graph touches the x -axis at exactly one point, the vertex, the function has **one repeated real zero**.

One intersection

$\Rightarrow$ one repeated real zero

No Real Zeros

If the graph does not intersect the x -axis, the function has **no real zeros**.

No intersection

$\Rightarrow$ no real zeros

Examples

Example 1

Graph

$$y = (x - 2)(x - 5).$$

From the graph, the x -intercepts are

$$x = 2, \quad x = 5.$$

Therefore, the zeros are 2 and 5.

Example 2

Graph

$$y = (x - 3)^2.$$

The graph touches the x -axis at

$$x = 3.$$

Therefore, the repeated real zero is 3.

Example 3

Graph

$$y = x^2 - 4.$$

The graph intersects the x -axis at

$$x = -2, \quad x = 2.$$

Therefore, the zeros are -2 and 2 .

Practice Exercises

Practice 1

Graph the quadratic function. Then determine its zero(s) from the graph.

$$y = (x - 2)(x - 5)$$

Practice 2

Graph the quadratic function. Then determine its zero(s) from the graph.

$$y = (x + 4)(x - 1)$$

Practice 3

Graph the quadratic function. Then determine its zero(s) from the graph.

$$y = (x - 3)^2$$

Practice 4

Graph the quadratic function. Then determine its zero(s) from the graph.

$$y = x^2 - 5x + 6$$

Practice 5

Graph the quadratic function. Then determine its zero(s) from the graph.

$$y = x^2 + x - 6$$

Practice 6

Graph the quadratic function. Then determine its zero(s) from the graph.

$$y = x^2 - 4$$

Practice 7

Graph the quadratic function. Then determine its zero(s) from the graph.

$$y = x^2 - 2x - 3$$

Practice 8

Graph the quadratic function. Then determine its zero(s) from the graph.

$$y = x^2 + 2x + 1$$

Practice 9

Graph the quadratic function. Then determine its zero(s) from the graph.

$$y = x^2 - 6x + 8$$

Practice 10

Graph the quadratic function. Then determine its zero(s) from the graph.

$$y = x^2 + 5x + 4$$

Activities

Activity 1

Graph the quadratic function. Then determine its zero(s) from the graph.

$$y = (x - 1)(x - 6)$$

Activity 2

Graph the quadratic function. Then determine its zero(s) from the graph.

$$y = (x + 2)(x - 5)$$

Activity 3

Graph the quadratic function. Then determine its zero(s) from the graph.

$$y = (x + 3)^2$$

Activity 4

Graph the quadratic function. Then determine its zero(s) from the graph.

$$y = x^2 - 7x + 12$$

Activity 5

Graph the quadratic function. Then determine its zero(s) from the graph.

$$y = x^2 - x - 6$$

Activity 6

Graph the quadratic function. Then determine its zero(s) from the graph.

$$y = x^2 - 9$$

Activity 7

Graph the quadratic function. Then determine its zero(s) from the graph.

$$y = x^2 + x - 6$$

Activity 8

Graph the quadratic function. Then determine its zero(s) from the graph.

$$y = x^2 - 6x + 9$$

Activity 9

Graph the quadratic function. Then determine its zero(s) from the graph.

$$y = x^2 - 8x + 15$$

Activity 10

Graph the quadratic function. Then determine its zero(s) from the graph.

$$y = x^2 + 7x + 10$$

Thank you!